

Anaiah Quinn

Gabriel Trevino

Armando Cabrera

Angel Bautista

Miguel Renteria

1. Assumptions

A cost-driving assumption is a statement that would materially change the estimate if it turned out to be wrong. Each assumption has an owner responsible for monitoring it.

Assumption	Why It Drives Cost	Owner
Seven-month build; two-year benefit horizon	Sets team size, delivery duration, and how many years of operating cost are included.	Sponsor
Apple and Android support from one shared codebase	Separate native applications would increase development, testing, and maintenance effort.	Technical Lead
Initial launch is limited to one regional/local market	Expanding geography increases vendor data collection, onboarding, quality checks, support, and possible API usage.	Sponsor / Product Owner
Vendor onboarding is limited at initial launch	Supporting hundreds or thousands of businesses increases database, moderation, onboarding, support, and integration requirements.	Project Manager
Peak usage is approximately 5,000 daily active users and up to 500 concurrent users	Determines hosting capacity, database sizing, performance testing, and infrastructure cost.	Technical Lead

2. Work packages

The current 4,045.2-hour estimate is retained, but it is distributed across development, QA, UX, and DevOps so the work breakdown is consistent with the project's actual delivery responsibilities. (674.2 * 6 = 4045.2)

Work Package	Role	Hours
Cross-platform customer app, authentication, event creation & dashboard	Developers	1,100
Backend, database, accounts & application APIs	Developers	850
Local vendor search, location services & vendor data	Developers	450

Work Package	Role	Hours
Invitations, RSVP workflow & attendance-related planning functionality	Developers	400
Testing and quality assurance	QA	600
User research, interface design & UX	UX	300
Environments, CI/CD, deployment & launch	DevOps	345.2
Total Delivery Effort		4,045.2

3. Effort Estimate

PERT Formula: $E = (O + 4M + P) \div 6$

Method / Input	Estimate
Bottom-up work package estimate	4,045.2 hours
Optimistic (O)	3,400 hours
Most likely (M)	4,000 hours
Pessimistic (P)	5,200 hours
PERT expected value	4,100 hours

4. Convert Hours to FTE- Months

The model uses 130 productive project hours per FTE-month. Project management is then added as one full-time PM across the seven-month build.

Role	Hours	Conversion	FTE-Months
Developers	2,800.0	$\div 130$	21.54
Quality Assurance	600.0	$\div 130$	4.62
User Experience	300.0	$\div 130$	2.31
DevOps	345.2	$\div 130$	2.66
Delivery subtotal	4,045.2	$\div 130$	31.13
Project Manager		1 FTE $\times$ 7 months	7.00
Total			38.13

5. Building the Staffing Plan

Each row is a role and each column is a month (M1 to M6). A cell is the fraction of one full-time person that month. 0.5 means half of one person's time.

Role	M1	M2	M3	M4	M5	M6	M7	FTE-Months
Project Manager	1	1	1	1	1	1	1	7.00
Developers	2	3	3.5	3.5	3.5	3.5	2.54	21.54
Quality Assurance	0.25	0.25	0.5	0.75	0.9	1	0.97	4.62
User Experience	1	0.75	0.3	0.15	0.11	0	0	2.31
DevOps	0.7	0.35	0.25	0.25	0.25	0.35	0.51	2.66
Total FTE	4.95	5.35	5.55	5.65	5.76	5.85	5.02	38.13

6. Calculating the loaded labor rate

Role	FTE-Months	Rate	Cost
Project Manager	7.00	\$13,000	\$91,000
Developers	21.54	\$13,000	\$280,020
Quality Assurance	4.62	\$13,000	\$60,060
User Experience	2.31	\$13,000	\$30,030
DevOps	2.66	\$13,000	\$34,580
Labor Total	38.13		\$495,690

7. Calculating Non-Labor Costs

Item	How Computed	Cost
Development/testing cloud infrastructure	\$2,000/month × 7	\$14,000
Development tools & licenses	\$8,000 setup + \$750 × 6	\$12,500
Maps/location/vendor-data API allowance	\$2,000/month × 7	\$14,000
Email/SMS/invitation notification services	\$1,000/month × 7	\$7,000
App-store/device/launch testing	One-time estimate	\$5,000
Security/privacy testing allowance	One-time estimate	\$12,500
Non-Labor Total		\$65,000

8. Price the cost of Ownership

Annual Operating Cost	How Computed	Per Year
Support & maintenance	0.75 FTE × \$13,000 × 12	\$117,000
Production hosting / database / storage	\$2,500/month × 12	\$30,000

Annual Operating Cost	How Computed	Per Year
Maps / vendor-data APIs	\$1,500/month × 12	\$18,000
Email / SMS / notifications	\$750/month × 12	\$9,000
Monitoring / licenses / services	\$1,000/month × 12	\$12,000
Operating Total		\$186,000 / year

9. Price the risks Contingency Reserve

Risk	Chance	Impact	EMV
Vendor/location/API integration is more complex than assumed	40%	\$50,000	\$20,000
Scope expands to real-time booking, payments, or additional vendor functionality	30%	\$75,000	\$22,500
Launch requires significant scaling, app-store, security, or deployment rework	25%	\$45,000	\$11,250
Contingency Reserve			\$53,750

Base build estimate	\$560,690
+ Contingency reserve	\$53,750
Cost baseline	\$614,440
Management reserve (5%)	\$30,722

10. Price the risks Contingency Reserve

Time-Phase the Budget

Month	Labor	Non-Labor	Total	Cumulative
M1	\$64,350	\$18,000	\$82,350	\$82,350
M2	\$69,550	\$7,000	\$76,550	\$158,900
M3	\$72,150	\$7,000	\$79,150	\$238,050
M4	\$73,450	\$7,000	\$80,450	\$318,500
M5	\$74,880	\$7,000	\$81,880	\$400,380
M6	\$76,050	\$11,000	\$87,050	\$487,430
M7	\$65,260	\$8,000	\$73,260	\$560,690

The Full Budget Request

Budget Line	Amount
Labor	\$495,690
Non-labor	\$65,000
Base build estimate	\$560,690
Contingency reserve (EMV)	\$53,750
Cost baseline	\$614,440
Management reserve (5%)	\$30,722
Total build budget request	\$645,162
Annual operating cost (separate line)	\$186,000 / year

Five Sanity Checks

Check	Festivo Result	Status
Totals match	Step 5 totals 38.13 FTE-months, matching Step 4.	PASS
Labor share is plausible	$\$495,690 \div \$560,690 = 88.4\%$. Software delivery is primarily labor.	PASS
Effort estimates agree	4,045.2 bottom-up hours vs. 4,100 PERT hours = approximately 1.4% difference.	PASS
Reserves trace to risks	\$53,750 of contingency is tied directly to three named risks.	PASS
Top-down comparison	Compare the \$645,162 build request to any sponsor/team target budget and explain the gap.	PENDING